DATA ITEM DESCRIPTION

Title: CONTROL SURFACE MASS PROPERTIES, FREEPLAY, AND RIGIDITY TESTS REPORT

Number: DI-SESS-82488

Approval Date: 20250708

AMSC Number: F10574

Limitation: N/A

DTIC Applicable: No

GIDEP Applicable: No

Preparing Activity: 11 (AFLCMC/EZFS)

Project Number: SESS-2025-015

Applicable Forms: N/A

Use/Relationship: The Control Surface Mass Properties, Freeplay, and Rigidity Tests Report is used to provide the data for the detailed evaluation of the mass and stiffness of the air vehicle's control surfaces and actuation system. It is used to establish free-play limits and update the aeroelastic and aeroservoelastic flutter analyses. The principal use of the report is to validate or update analytical models.

a. This Data Item Description (DID) is applicable to any contract for which an air vehicle is being developed, or a control surface, its associated backup structure, or linkages are being modified.

b. This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in section 5 of MIL-STD-1530, *Aircraft Structural Integrity Program (ASIP)* (copies of this document are available online at https://quicksearch.dla.mil).

c. This DID supersedes DI-MISC-81387, *Airframe Rigidity Test Report* (copies of this document are available online at https://quicksearch.dla.mil).

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Control Surface Mass Properties, Freeplay, and Rigidity Tests Report shall be in the contractor's format.

3. Content. The Control Surface Mass Properties, Freeplay, and Rigidity Tests Report shall describe the contractor's analyses and ground test activities associated with characterizing the mass, inertia, rigidity, freeplay, and wear rates of the flight control augmentation system and its components. The report shall contain the results and findings of the tests and measurements used to validate and update the aeroelastic and aeroservoelastic models and analyses, to include the following:

a. The report shall contain the planning, procedures, results, and findings for the following tests and measurements:

   (1) Control surface, tab, and actuator rigidity tests.

   (2) Control surface and tab freeplay tests.

   (3) Control surface, tab, and actuator component wear tests.

   (4) Mass property measurements of control surfaces and tabs.

   (5) Flutter damper qualification and wear tests.

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b. Presentation of test plans and results, which shall include:

(1) Descriptions and photographs of the air vehicle, fixturing, rigging, ground equipment, instrumentation, and component(s) under test.

(2) Descriptions of test procedures, test conditions, test methods, test equipment, load application method and locations, loading rates, data acquisition system, and methods of data reduction analysis.

(3) Copy of the test matrix as planned to be executed.

(4) Description of the methods used for selecting test instrumentation types, test instrumentation measurement capabilities, and test measurement locations.

(5) Description of the methods used for mounting instrumentation.

(6) Specifications for all test instrumentation.

(7) Descriptions of the aircraft flight control and actuation system including layout, dimensions, and method of operation.

(8) Description of the methods used to control actuators and actuation systems.

(9) Defined measurement units including axis labels of plots and figures.

(10) Plots of the transfer functions versus frequency for vibration sensors.

(11) Measurements of the freeplay and loop stiffness of each control surface and tab of the flight control augmentation system.

(12) Freeplay contribution of the individual components of the flight control augmentation system linkage and total freeplay for each control surface and tab along with a description of the methodology used to interpret the test data.

(13) Wear rates of individual components of the flight control augmentation system linkage for each control surface and tab along with a description of the methodology used to interpret the test data.

(14) Actuator impedance results as a function of frequency.

(15) Flutter damper impedance and functional service life limits.

(16) Mass and inertial properties of the control surfaces and tabs.

(17) Copy of the test log as executed.

(18) Control surface and tab balance weight locations, tolerances, and configurations, including any necessary adjustments to the balance weight based on test and measurement results.

End of DI-SESS-82488

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